C5.4 How are the amounts of chemicals in solution measured?

Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Quantitative analysis is used by chemists to make measurements and calculations to show the amounts of each component in a sample. **Concentrations sometimes use the units g/dm^3 but more often are expressed using moles, with the units mol/dm^3. Expressing concentration using moles is more useful because it links more easily to the reacting ratios in the equation.** The concentration of acids and alkalis can be analysed using titrations. Alkalis neutralise acids. An indicator is used to identify the point when neutralisation is just reached. During the reaction, hydrogen ions from the acid react with hydroxide ions from the alkali to form water. The reaction can be represented using the equation $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$ As with all quantitative analysis techniques, titrations follow a standard procedure to ensure that the data is collected safely and is of high quality, including selecting samples, making rough and multiple repeat readings and using equipment of an appropriate precision (such as a burette and pipette).	1. identify the difference between qualitative and quantitative analysis *(separate science only)*	**Specification links:** • Strong and weak acid chemistry (C6.1) **Practical Work:** • Making up a standard solution.
	2. explain how the mass of a solute and the volume of the solution is related to the concentration of the solution and calculate concentration using the formula: $\textbf{concentration (g/dm}^3\textbf{)} = \dfrac{\textbf{mass of solute (g)}}{\textbf{volume (dm}^3\textbf{)}}$ M3b, M3c	
	3. explain how the concentration of a solution in mol/dm^3 is related to the mass of the solute and the volume of the solution and calculate the molar concentration using the formula $\textbf{concentration (mol/dm}^3\textbf{)} = \dfrac{\textbf{number of moles of solute}}{\textbf{volume (dm}^3\textbf{)}}$ M3b, M3c	**Practical work** • Acid-base titrations. • Use of appropriate measuring apparatus, measuring pH, use of a volumetric flask to make a standard solution, titrations using burettes and pipettes, use of acid-base indicators, safe handling of chemicals.
	4. describe neutralisation as acid reacting with alkali to form a salt plus water including the common laboratory acids hydrochloric acid, nitric acid and sulfuric acid and the common alkalis, the hydroxides of sodium, potassium and calcium	

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Data from titrations can be assessed in terms of its accuracy, precision and validity. An initial rough measurement is used as an estimate and titrations are repeated until a level of confidence can be placed in the data; the readings must be close together with a narrow range. The true value of a titration measurement can be estimated by discarding roughs and taking a mean of the results which are in close agreement. The results of a titration and the equation for the reaction are used to work out the concentration of an unknown acid or alkali.	5.	recall that acids form hydrogen ions when they dissolve in water and solutions of alkalis contain hydroxide ions
	6.	recognise that aqueous neutralisation reactions can be generalised to hydrogen ions reacting with hydroxide ions to form water
	7.	describe and explain the procedure for a titration to give precise, accurate, valid and repeatable results *PAG6*
	8.	Evaluate the quality of data from titrations
	9.	**explain the relationship between the volume of a solution of known concentration of a substance and the volume or concentration of another substance that react completely together *(separate science only)***

Linked learning opportunities
Ideas about Science: • Justify a technique in terms of precision, accuracy and validity of data to be collected, minimising risk. Use of range and mean when processing titration results, analysis of data. (IaS1, IaS2)